Phase 1 Report: Study, Data, and Descriptive Statistics

Living situation, executive function, and frontal cortical thickness in older adults: a multimodal (UDS + MRI) analysis of an NACC subset

Statistics Project, Fall 1405 (2026) | Track C (NACC, MRI + UDS) | Phase 1 | Hasti Mirshamsi, student ID 403100281

Main question (PROJECT_CARD.md). Is living situation (`NACCLIVS`) associated with executive function (`TRAILB`) and with the cortical thickness of selected frontal regions in older adults? This phase is purely descriptive: it documents where the data come from, what a row means, how UDS and MRI were linked, how good the data are, and how the main variables are distributed. No hypothesis tests are performed.

1. Data source and collection mechanism

Source. The data are a subset of the National Alzheimer's Coordinating Center (NACC) database that the teaching team prepared for Track C; the files were received from the teaching assistants on 10 October 2026. According to the NACC website, the database holds data on more than 56,000 participants collected over 25+ years at more than 40 Alzheimer's Disease Research Centers (ADRCs) funded by the National Institute on Aging, and is available to researchers on request. The **UDS** (Uniform Data Set) component contains standardized clinical information (demographics, medical history, medications, functional status, neuropsychological tests, clinical diagnosis); the **MRI** component contains regional brain measures for the participants in the subset.

Files used. `udslong_imputed.csv` (1237 rows x 180 columns) and `mriscanclean.csv` (1237 rows x 126 columns). A third file (`combined_adjacency_matrix.csv`, a region-by-region adjacency matrix) is not used in this project.

Type of study. The data come from clinical research centres, not from a probability survey and not from an experiment: this is an **observational** data set. Participants were not assigned to living situations, so any association we later find describes who lives how and how they perform, and cannot be read as an effect of living situation.

Access limits. The raw files were provided by the teaching team and are not redistributed in the repository (see `data/README.md`). The file name says *imputed*, and decimals in whole-number variables suggest prior imputation or transformation by an unknown procedure (Section 6). The criteria that selected these 1,237 participants out of the full NACC database are not documented in the files; they must be treated as unknown.

2. Unit of observation and data structure

One row = one participant (`NACCID`). Both files contain exactly one row per participant:

rows_per_participant	n_participants
1	1237

Table 1. Rows per participant in the UDS file (the MRI file has the same structure).

Structure: UDS file (1,237 x 180) <-- `NACCID`, one-to-one --> MRI file (1,237 x 126). There is no hierarchy inside the delivered files (no repeated visit rows). However, the UDS is longitudinal by design and the file name (*udslong*) suggests it was derived from a long file with several visits per person. Neither file has a visit number or date, so **we cannot count visits per participant or say which visit was kept**. **Rule for the following phases:** the single delivered record per participant is the independent observation (so each person enters every analysis once). Which visit this record represents, and how close in time the UDS and MRI are, will be requested from the teaching team; until then this is stated as an open limitation.

3. Sampling, selection, and generalisability

ADRC participants are volunteers who come to specialised centres, many through memory-clinic referral, community outreach, or family history. The sample is therefore **not random** and not representative of older adults in general. The data themselves show typical features of such a sample (sample A: aged 65+ with a valid living situation, n = 861; 864 participants are 65+ before excluding the 3 with NACCLIVS = 5 or 9): median education 16

years, 57.1% women, 66.7% living with a spouse or partner, and 65.5% with normal cognition (Table 8a). Only people who have an MRI are included, which may add another layer of selection (people able and willing to be scanned). **Interpretable population:** NACC participants of this subset who are aged 65+ and have the required data. Statements about older adults in general are not supported.

4. Variable map and classification (C.1, task 4)

The complete maps are in `reports/phase1/tables/variable_map_uds.csv` and `variable_map_mri.csv`. Table 2 summarises the UDS variables by domain; the *scale* of each column was inferred from its values and must be confirmed in the RDD (variable guide).

domain	subdomain	n variables
cognitive	neuropsychological tests	10
demographic	demographics	11
diagnostic	cognitive status	1
functional	independence / CDR / FAQ	19
medical	behavioural symptoms (NPI-Q)	25
medical	cardio/cerebrovascular	11
medical	family history	11
medical	genetic	1
medical	medications	21
medical	metabolic / other conditions	16
medical	mood (GDS)	17
medical	neurological	8
medical	physical measures / sensory	12
medical	psychiatric history	9
medical	substance use	7

Table 2. UDS variables by domain and sub-domain (179 variables besides NACCID). 'Medical' also holds behavioural (NPI-Q) and mood (GDS) items.

MRI. 124 columns = 31 cortical regions x 2 hemispheres (LH/RH) x 2 measures: `GVOL` (gray-matter volume, mm^3) and `AVGTH` (average cortical thickness, mm); these units were provided for this data set and are to be cross-checked against the RDD. The region names (e.g. CAUDALMIDDLEFRONTAL, ENTORHINAL) follow a FreeSurfer-style cortical parcellation; units and processing details must be confirmed in the RDD. By lobe: 10 frontal, 5 parietal, 7 temporal, 4 occipital, 4 cingulate and 1 insular region.

variable	source	meaning	type / scale	role	summary / plot choice
NACCLIVS	UDS	Living situation	Nominal (categorical, 4 usable groups)	Main explanatory / grouping variable	Frequency table, bar chart; ANOVA/chi-square family later
TRAILB	UDS	Trail Making B time (s)	Quantitative continuous (ratio); right-skewed, capped at 300 s, special codes	Cognitive response (executive function)	Median/IQR, boxplot, log scale
NACCAGE	UDS	Age (years)	Quantitative continuous	Adjusting variable / sample definition (>=65)	Mean/SD or median/IQR, histogram
SEX	UDS	Sex	Binary	Adjusting variable	Frequency table
EDUC	UDS	Years of education	Quantitative discrete	Adjusting variable	Median/IQR (ceiling near 16-20 years)
FRONTAL_THK	MRI (derived)	Mean AVGTH of 6 frontal values (caudal/rostral middle frontal, superior frontal; LH+RH), mm	Quantitative continuous	Imaging response	Mean/SD, histogram, boxplot

Table 3. Main variables, scale of measurement, role, and the summaries/plots this classification implies.

Frontal thickness composite. FRONTAL_THK is the simple mean of six values: AVGTH of caudal middle frontal, rostral middle frontal and superior frontal regions in the left and right hemispheres. The left and right means correlate at r = 0.85 in the analytic sample, which supports combining them. The regions were fixed in the project

card before looking at any group comparison.

5. Linking UDS and MRI (C.3) and analytic sample

step	n
UDS file: records	1237
UDS file: unique participants (NACCID)	1237
MRI file: records	1237
MRI file: unique participants (NACCID)	1237
Participants in both files	1237
Only in UDS (no MRI match)	0
Only in MRI (no UDS match)	0
Duplicate NACCID in UDS / in MRI	0 / 0

Table 4. Linkage of the two files by NACCID.

The two files contain the same 1,237 participants, with no unmatched records and no duplicate IDs. The diagnosis variable NACCUDSD, present in both files, is identical for every participant after linking (a useful check that the link is correct). **All 1,237 delivered participants therefore have both modalities.**

step	n
Linked participants (UDS + MRI)	1237
Age >= 65	864
Valid NACCLIVS in {1,2,3,4} (analytic sample A)	861
A with observed TRAILB (primary definition)	798
A with TRAILB incl. imputed in-range values (sensitivity)	821
A with frontal thickness (all 6 regions)	861
A with valid SEX, EDUC and age	860
A with TRAILB (primary) + thickness + SEX/EDUC/age (complete case)	797

Table 5. Flow from the linked data to the analytic sample A (age >= 65, valid NACCLIVS in 1-4) and availability of the main variables.

Exclusions from the linked data: 373 participants younger than 65 (age range of the file 41-91); 1 participant with NACCLIVS = 5 (other), too small to analyse as a group; 2 with NACCLIVS = 9 (unknown). The 65+ threshold is the pre-declared definition of 'older adults' in the project card (a threshold of 60 will be used as a sensitivity analysis in Phase 5).

6. Data-quality audit (task 5)

Expected ranges and special codes below follow NACC conventions and should be double-checked against the RDD; the code only counts and flags and never deletes participants.

variable	expected_range	special_codes	n_nan	n special code	n non-integer	n non-int. out of range	n int. out of range
SEX	1-2	-	0	0	0	0	0
EDUC	0-36	[99]	0	2	0	0	0
NACCAGE	18-110	-	0	0	0	0	0
NACCLIVS	1-5	[9]	0	3	0	0	0
MARISTAT	1-6	[9]	0	3	0	0	0
NACCUDSD	1-4	-	0	0	0	0	0
TRAILB	0-300	[995, 996, 997, 998]	0	55	48	9	0
TRAILA	0-150	[995, 996, 997, 998]	0	21	48	4	1
MOCATOTS	0-30	[88]	0	4	249	5	0
NACCAPOE	1-6	[9]	0	278	0	0	0

Table 6a. Audit of the variables used in the project (+ MARISTAT, MOCATOTS, NACCAPOE).

Missing values. There are no NaN cells in either file (UDS 0, MRI 0), but this does not mean complete data: unknown/not-done answers are stored as special codes (for example 995-998 for the trail-making tests, 99 for education, 9 for living situation, 278 participants with APOE = 9).

Imputation. Questionnaire and test variables that should only contain whole numbers contain decimals: **137 columns, 5,311 cells, 753 of 1237 participants** have at least one such value (table_suspected_imputed_columns.csv). **Suspected prior imputation or transformation; the exact procedure is unknown.** Decimals in whole-number variables do not prove a specific method. Some of these values are impossible for the instrument (e.g. a Trails B time of 523.6 s when the test stops at 300 s). The method, whether MRI values were also processed, and whether the file is the final analysis-ready version must be asked from the teaching team (Section 13).

value class	TRAILB n	TRAILA n	TRAILB %	TRAILA %
observed	1079	1152	87.2	93.1
observed_at_cap	55	15	4.4	1.2
special_code	55	21	4.4	1.7
imputed_in_range	39	44	3.2	3.6
imputed_out_of_range	9	4	0.7	0.3
integer_out_of_range	0	1	0	0.1

Table 6b. Classification of all Trail Making values in the file (TRAILB limit 300 s, TRAILA limit 150 s).

TRAILB. 87.2% of TRAILB values are ordinary observed times; 4.4% are at the 300 s limit (true time is at least 300 s: *censored*); 4.4% are special codes (test not completed: physical, cognitive or other problem, or refusal); and 3.9% are non-integer values judged to be imputed (9 beyond 300 s). In sample A, 798 of 861 participants have an observed TRAILB (**92.7%**); allowing in-range imputed values would add 23. The primary definition uses observed values only; the version with in-range imputed values is kept for the Phase 5 sensitivity analysis.

check	n	comment
NACCUDSD = 1 (normal) and CDRGLOB >= 1	0	contradictory
NACCUDSD = 4 (dementia) and CDRGLOB = 0	0	contradictory
NACCUDSD = 1 (normal) and CDRGLOB = 0.5	75	possible (clinician judgement), not an error
NACCLIVS = 1 (alone) and MARISTAT in {married, domestic partner}	17	plausible (partner lives elsewhere); flagged
NACCLIVS = 2 (spouse/partner) and MARISTAT in {widowed, divorced, never married}	14	plausible (unmarried partner); flagged
TRAILB < TRAILA (both observed)	2	unusual (part B is normally slower)
EDUC (valid) > NACCAGE - 3	0	impossible
NACCAGE < 18 or > 110	0	impossible

Table 6c. Logical-consistency checks.

No contradictory diagnosis/CDR pairs and no impossible ages or education were found. The flagged living situation vs marital status combinations (for example living alone while married or in a domestic partnership) are plausible and kept.

MRI. No NaN. Volumes: 2 zero values (LH entorhinal, LH posterior cingulate), not used in this project. Thickness below 1 mm is biologically unlikely and probably a segmentation error: 58 such cells in 6 participants across all regions; in the six frontal values used here, 7 cells in 3 participants (2 in sample A). They are flagged (FRONTAL_flag_lt1mm), not deleted; their influence will be examined in Phases 4-5.

problem	how many	decision	reason
Special codes (EDUC 99, NACCLIVS 9, TRAILB/TRAILA 995-998)	n per variable: see table_audit_main_variables	Recode to NaN on a working copy	These are 'unknown/not done' codes, not measurements; averaging them would be meaningless.
Non-integer (suspected imputed) TRAILB/TRAILA values	TRAILB: 48 values	Excluded from primary TRAILB; kept in sensitivity definition (in-range only)	Imputation method unknown; values > 300 s are impossible for this test.
TRAILB = 300 s	55 participants	Kept; flagged as censored at the time limit	True time is >= 300 s; Phase 3/4 will check sensitivity (e.g. log scale, rank methods).
Other suspected imputed columns	137 columns, 5311 cells	Not used in the main question; documented only	Variables are outside the analysis plan; ask teaching team about imputation method.
MRI thickness < 1 mm in a frontal region	7 cells / 3 participants	Kept and flagged (FRONTAL_flag_lt1mm); influence checked in Phase 4/5	Biologically unlikely (likely segmentation error) but no rule allows silent deletion.
MRI volume = 0	2 cells	Kept; not used in the main question	Volumes are not part of the analysis plan.
Duplicate NACCID	0	None needed	One record per participant in both files.
Logical inconsistencies	see table_logic_checks	No row removed; flagged items are plausible	No contradictory NACCUDSD/CDR or impossible age/education combinations.

Table 6d. Decision for every data problem and the reason.

coding used in this report	status	source / note
NACCLIVS: 1 alone, 2 spouse/partner, 3 relative/friend, 4 group, 5 other, 9 unknown	Verified	Variable page of the RDD provided for NACCLIVS (derived variable, numeric longitudinal).
NACCUDSD: 1 normal, 2 impaired-not-MCI, 3 MCI, 4 dementia	Verified (secondary check)	NACC documentation lists NACCUDSD as 'Cognitive status at UDS visit'; the four labels and numeric order match a published coding supplement. Confirm order in the RDD.
TRAILB: 0-300 s, 300 = not finished by 300 s; 995 physical, 996 cognitive/behaviour, 997 other problem, 998 verbal refusal	Verified	Official NACC UDS Form C2 (and C1) instructions.
TRAILA: 0-150 s, 150 = not finished by 150 s; same codes 995-998; TRAILARR/TRAILALI = commission errors (0-40) / correct lines (0-24)	Verified	Official NACC UDS Form C2 (and C1) instructions.
SEX (1 male, 2 female), EDUC (0-36, 99 unknown), MARISTAT labels, MOCATOTS (0-30, 88), NACCAPOE (9 missing)	Not yet verified	Based on NACC conventions; remain a limitation until checked in the RDD.
MRI units: AVGTH in mm, GVOL in mm^3	Provided, to cross-check	Units provided for this data set; to be confirmed in the RDD.

Table 6e. Verification status of the variable coding used in this report.

7. Missing data and selection of the multimodal sample (C.4)

variable	n (all 1,237)	% (all)	n (A, 861)	% (A)
NACCLIVS (valid code 1-5)	3	0.2	0	0
NACCAGE	0	0	0	0
SEX	0	0	0	0
EDUC (99 = unknown)	2	0.2	1	0.1
TRAILB (primary: observed only)	103	8.3	63	7.3
TRAILB (incl. in-range imputed)	64	5.2	40	4.6
Frontal thickness (6 regions)	0	0	0	0
All 124 MRI columns (any NaN)	0	0	0	0

Table 7a. Unavailable values in the main variables (NaN, special code, or not usable under the primary definition).

Only TRAILB has a noticeable amount of unavailable data. In the MRI file nothing is missing. **We cannot compare participants with and without MRI**, because both delivered files describe the same 1,237 participants (the UDS file is also restricted to people with MRI), so there is no UDS-only comparison group; the possible selection into the MRI subset therefore remains a limitation we can describe but not measure. What we can examine is who has an

unusable TRAILB:

(sample A)	TRAILB not usable	TRAILB usable
n	63	798
age_mean	72.73	72.19
age_sd	5.58	5.29
pct_female	60.32	56.89
educ_median	16	16
pct_alone	15.87	23.18
pct_MCI	26.98	23.31
pct_dementia	44.44	7.52
frontal_thk_mean	2.36	2.38

Table 7b. Participants in A with and without a usable TRAILB (descriptive).

People without a usable TRAILB are similar in age, sex and education, but **44.4% of them have dementia versus 7.5% of those with a usable TRAILB**. The availability of TRAILB is therefore associated with the observed cognitive diagnosis, suggesting that the data are not Missing Completely At Random (MCAR). The missingness may be explainable by observed variables such as diagnosis (MAR), although MNAR mechanisms cannot be ruled out; this analysis cannot tell MAR from MNAR. Later results for TRAILB mostly describe participants able to complete the test.

8. Frequency tables, centre and spread (tasks 6-7)

variable	category	n	pct
NACCLIVS (sample A)	Alone	195	22.6
NACCLIVS (sample A)	Spouse/partner	574	66.7
NACCLIVS (sample A)	Relative/friend	56	6.5
NACCLIVS (sample A)	Group	36	4.2
SEX (sample A)	Female	492	57.1
SEX (sample A)	Male	369	42.9
NACCUDSD (sample A)	Normal	564	65.5
NACCUDSD (sample A)	Impaired-not-MCI	6	0.7
NACCUDSD (sample A)	MCI	203	23.6
NACCUDSD (sample A)	Dementia	88	10.2

Table 8a. Frequency tables for the sample A (percentages sum to 100 within each variable).

Reading the tables. About two thirds live with a spouse or partner (66.7%) and 22.6% live alone, while 'relative/friend' (n = 56) and 'group' (n = 36) are small; the 'other' category (n = 1) and 'unknown' (n = 2) could not be analysed. Women are 57.1% of the sample. Nearly one third are impaired in some way (23.6% MCI, 10.2% dementia); the 'impaired-not-MCI' group has only 6 people and cannot be analysed as a separate group.

variable	n	missing	mean	sd	min	Q1	median	Q3	max	IQR	skewness	Tukey outliers
Age (years)	861	0	72.23	5.31	65	68	71	76	91	8	0.69	6
Education (years)	860	1	16.22	2.89	0	14	16	18	25	4	-1.03	12
TRAILB, observed (s)	798	63	102.34	63.51	28	63	82	113.75	300	50.75	1.97	69
log10 TRAILB	798	63	1.95	0.22	1.45	1.8	1.91	2.06	2.48	0.26	0.77	49
Frontal thickness (mm)	861	0	2.38	0.12	1.03	2.31	2.39	2.45	2.8	0.14	-2.52	16
Frontal thickness LH (mm)	861	0	2.39	0.13	1.19	2.32	2.4	2.46	2.84	0.14	-1.93	16
Frontal thickness RH (mm)	861	0	2.37	0.13	0.88	2.31	2.38	2.44	2.75	0.14	-3.04	17

Table 8b. Summary of the quantitative variables (sample A). TRAILB is based on observed values only.

Which measure suits which variable. *Age* (mean 72.2, SD 5.3; median 71) is mildly right-skewed (skewness 0.69); mean/SD and median/IQR agree closely, either can be used. *Education* is left-skewed with a ceiling (median

16, IQR 4) so median/IQR is more appropriate. *TRAILB* is strongly right-skewed (skewness 1.97; mean 102 s versus median 82 s), with a pile-up at the 300 s limit; the median/IQR (63-114 s) describes a typical participant, and the log scale reduces skewness to 0.77, which is why a log transform will be considered in Phases 3-4. *Frontal thickness* is nearly symmetric in the centre (mean 2.38 mm, SD 0.12) but has a long left tail from a few implausibly low values (minimum 1.03 mm; skewness -2.52); mean/SD are sensitive to these, so median/IQR will be reported alongside.

9. Figures (task 8)

Figure 1. Living situation, sample A: aged 65+ with valid NACCLIVS (n = 861)

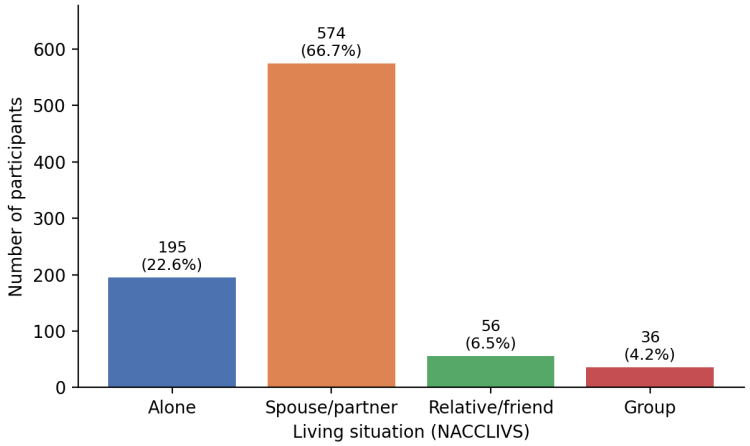


Figure 1. Counts and percentages of the four living-situation groups. The two small groups (relative/friend, group) limit any later comparison involving them.

Figure 2. TRAILB, observed values (n = 798; 46 at the 300 s limit)

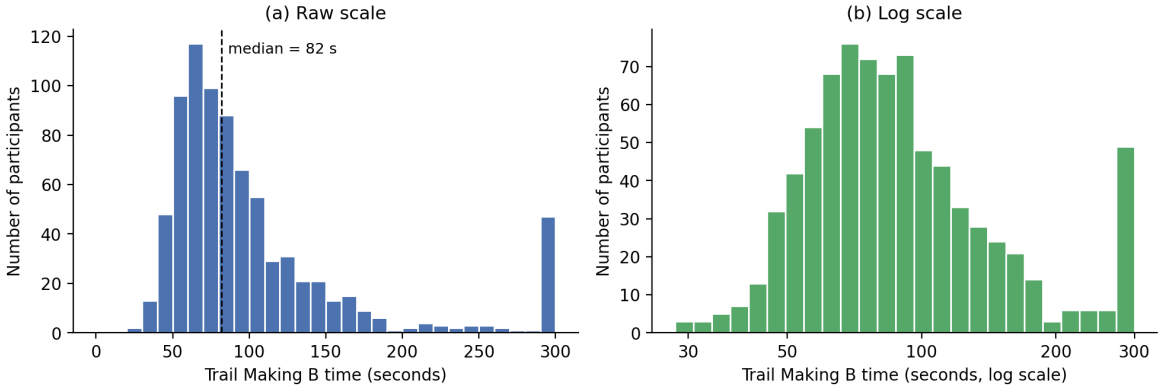


Figure 2. Distribution of observed TRAILB times. The spike at 300 s is the test time limit (censoring); the log scale (b) makes the shape more symmetric.

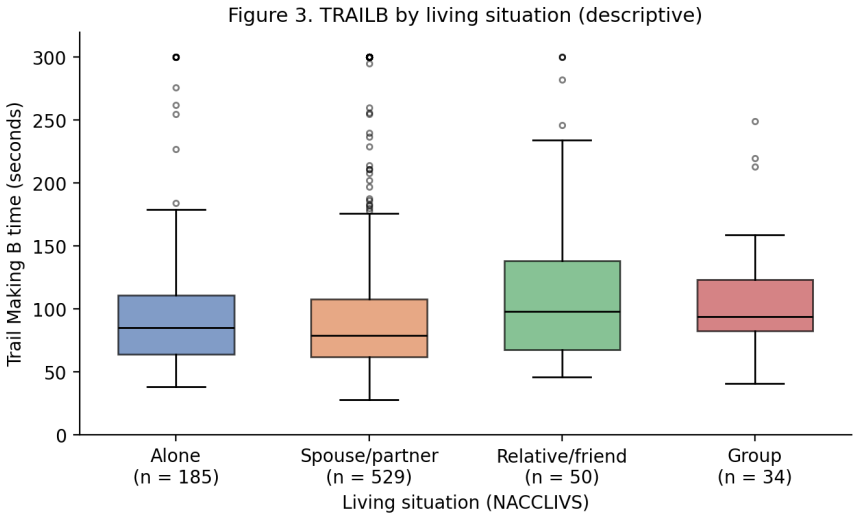


Figure 3. TRAILB by living situation. Boxes overlap strongly; many outliers are high times, several at the 300 s limit. Descriptive only.

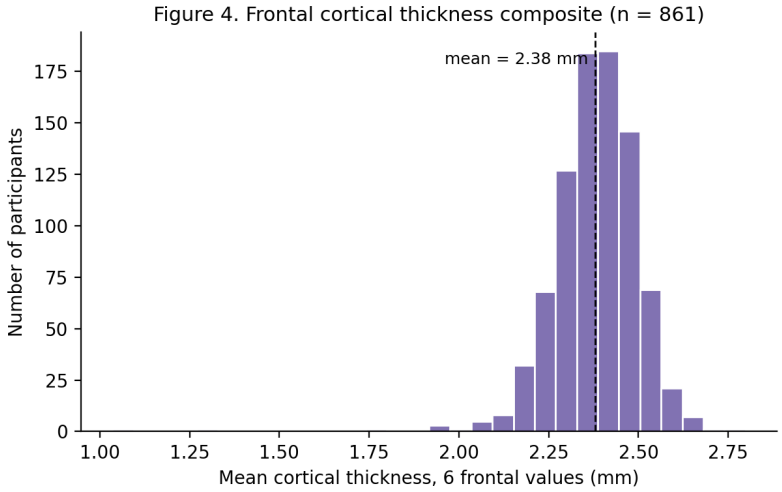


Figure 4. Frontal thickness composite (mean of 6 regional values). The x-axis extends to 1.0 mm because of a few implausibly low values (visible as isolated points in Figure 5).

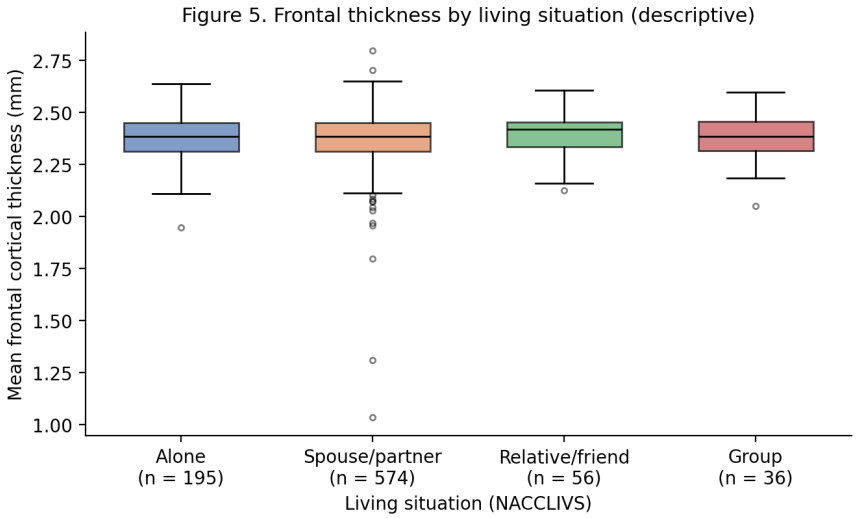


Figure 5. Frontal thickness by living situation. The y-axis does not start at 0 because the box position, not its length from zero, carries the information. The medians are nearly the same in all groups.

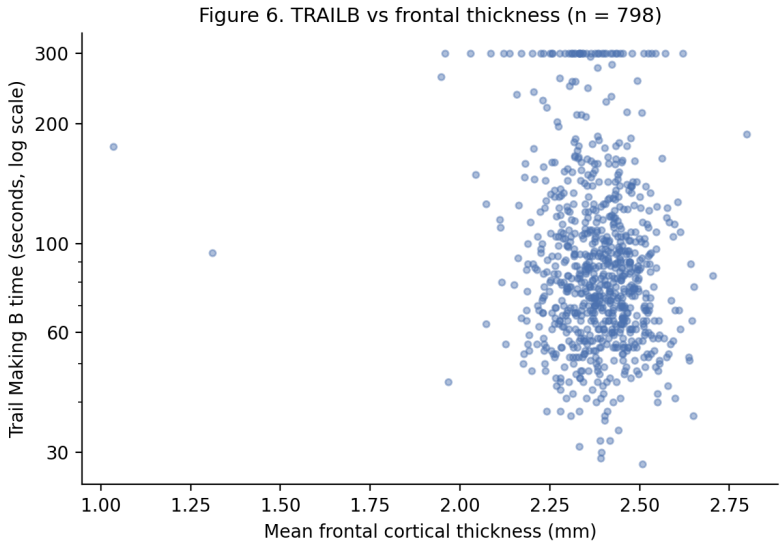


Figure 6. TRAILB (log scale) versus frontal thickness; each point is one participant (n = 798). The horizontal line of points at 300 s is the censoring limit. The two points far left are the implausibly thin values.

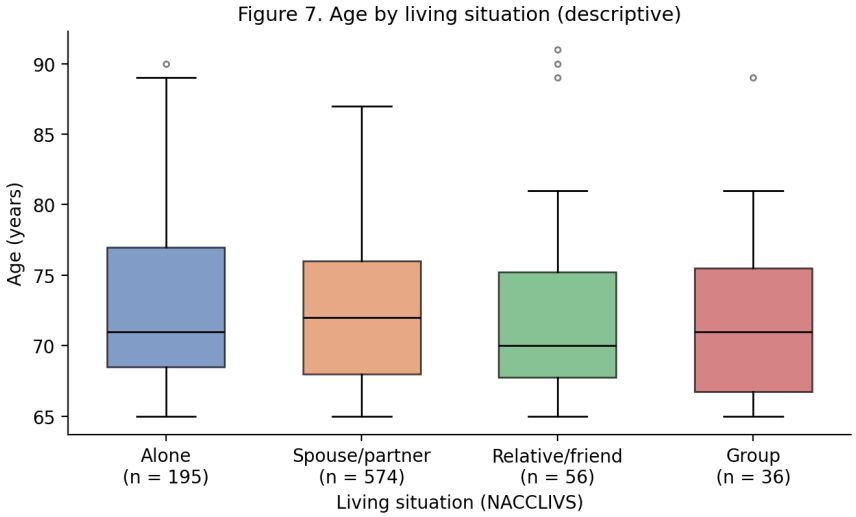


Figure 7. Age by living situation. The minimum is 65 by definition of sample A. Age is similar across groups.

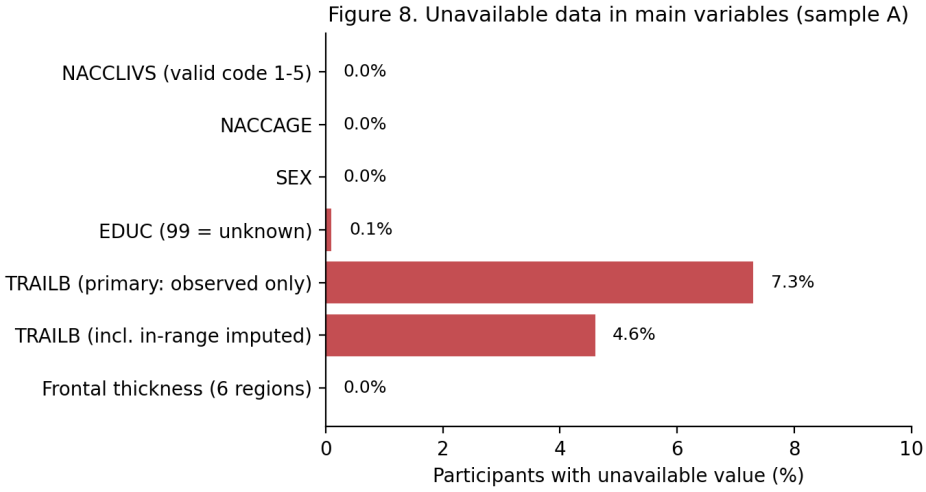


Figure 8. Percentage of participants in A with an unavailable value in each main variable.

10. Descriptive comparison across living situations (task 9)

	Alone	Spouse/partner	Relative/friend	Group
n	195	574	56	36
Age, mean (SD)	72.8 (5.8)	72.1 (5.0)	71.8 (5.9)	71.7 (5.8)
Women, %	78	48	80	53
MCI, %	18	26	18	25
Dementia, %	3	13	11	14
TRAILB: n with observed value	185	529	50	34
TRAILB median [Q1-Q3], s	85 [64-111]	79 [62-108]	98 [68-138]	94 [82-124]
TRAILB mean (SD), s	100 (60)	101 (65)	117 (69)	108 (46)
TRAILB at 300 s limit, %	4.9	6.4	6.0	0.0
Frontal thickness, mean (SD), mm	2.38 (0.11)	2.38 (0.13)	2.40 (0.09)	2.38 (0.12)

Table 10. Descriptive comparison of sample A by living situation (no tests, no inference).

What the numbers show. Median TRAILB time was 85 s (alone), 79 s (spouse/partner), 98 s (relative/friend) and 94 s (group), with large spread inside each group (interquartile ranges overlap heavily). The two small groups have the higher medians, but they contain only 50 and 34 participants with an observed TRAILB. Mean frontal thickness is 2.38-2.40 mm in all groups: practically identical in this sample.

Groups differ in who they contain. 78% of those living alone are women compared with 48% of those living with a spouse/partner; dementia is recorded for 2.6% of those living alone but 12.5% of those living with a spouse/partner. Sex, cognitive status and living situation are thus entangled. Any later difference in TRAILB between groups may reflect these other factors, which is why sex, age and education (and, in sensitivity analyses, diagnosis) matter in Phases 3-4. This phase does not claim that any difference exists in the population.

living situation	Normal	Impaired-not-MCI	MCI	Dementia	Total
Alone	154 (79.0%)	0 (0.0%)	36 (18.5%)	5 (2.6%)	195
Spouse/partner	348 (60.6%)	6 (1.0%)	148 (25.8%)	72 (12.5%)	574
Relative/friend	40 (71.4%)	0 (0.0%)	10 (17.9%)	6 (10.7%)	56
Group	22 (61.1%)	0 (0.0%)	9 (25.0%)	5 (13.9%)	36

Table 10b. Cognitive diagnosis (NACCUDSD) by living situation in sample A: number of participants (row percentage).

Why diagnosis matters for the research question. Cognitive status is distributed very differently across living situations: dementia is recorded for 2.6% of participants living alone versus 12.5% living with a spouse/partner, 10.7% with a relative/friend and 13.9% in a group setting (the 'impaired-not-MCI' category has too few people to interpret). Because diagnosis is also strongly related to whether TRAILB can be obtained (Table 7b), it could confound a living situation - TRAILB association, or lie on its pathway (for example, cognitive decline leading to a change in living arrangements). The project card keeps age, sex and education as the initial adjusting variables; diagnosis is planned as a stratification or sensitivity variable in later phases, and its role must be stated explicitly in any interpretation.

11. Joint cognitive and imaging summary (C.5)

variable	n	mean	sd	min	Q1	median	Q3	max	IQR	skewness	Tukey outliers
TRAILB (s)	798	102.34	63.51	28	63	82	113.75	300	50.75	1.97	69
Frontal thickness (mm)	798	2.38	0.12	1.03	2.32	2.39	2.45	2.8	0.14	-2.67	14

Table 11. TRAILB (UDS) and frontal thickness (MRI) in the 798 participants of A who have both.

TRAILB ranges from 28 to 300 s (median 82, IQR 51) and is strongly right-skewed with 69 Tukey outliers (46 of them the censored values at 300 s). Frontal thickness is tightly concentrated (mean 2.38 mm, SD 0.12) with 14 Tukey outliers (12 unusually thin, 2 unusually thick). The two measures live on different scales and have different uncertainty, which is why they are summarised separately here. Figure 6 shows no obvious pattern by eye; the covariance and correlation will be examined formally in Phase 2 and Phase 4, and nothing causal can be concluded from such a plot.

11b. Exploratory component: 12 frontal MRI measures (descriptive only)

Purpose and rules. Before any test, we looked at how the frontal MRI measures relate to each other, to age and education, and to the Trail Making tests. The 12 MRI measures are thickness (AVGTH) and gray-matter volume (GVOL) of caudal middle frontal, rostral middle frontal and superior frontal regions in both hemispheres. This part is **exploratory and descriptive**: Spearman rank correlations (robust to the skewness of TRAILA/TRAILB and to extreme thickness values) with pairwise-complete data, **no p-values**, and it was **not used to choose the confirmatory question**, which was fixed in the project card.

MRI variable	TRAILB	TRAILA	AGE	EDUC
LH caudal MF thick	-0.1	-0.09	-0.19	-0.09
LH rostral MF thick	-0.04	0	-0.04	-0.05
LH superior F thick	-0.08	-0.07	-0.14	-0.05
RH caudal MF thick	-0.12	-0.09	-0.11	-0.05
RH rostral MF thick	-0.08	-0.08	0	0.01
RH superior F thick	-0.09	-0.06	-0.06	0.01
LH caudal MF vol	-0.14	-0.19	-0.1	0.1
LH rostral MF vol	-0.14	-0.19	-0.11	0.13
LH superior F vol	-0.18	-0.19	-0.11	0.16
RH caudal MF vol	-0.15	-0.19	-0.12	0.07
RH rostral MF vol	-0.16	-0.21	-0.13	0.1
RH superior F vol	-0.16	-0.18	-0.1	0.14

Table 12. Spearman correlation of each frontal MRI measure with TRAILB, TRAILA, age and education (sample A; TRAILA/TRAILB observed values only).

Figure 9. Exploratory Spearman correlation matrix (sample A, pairwise-complete; no p-values)
thick = cortical thickness (mm), vol = gray-matter volume; MF = middle frontal, F = frontal

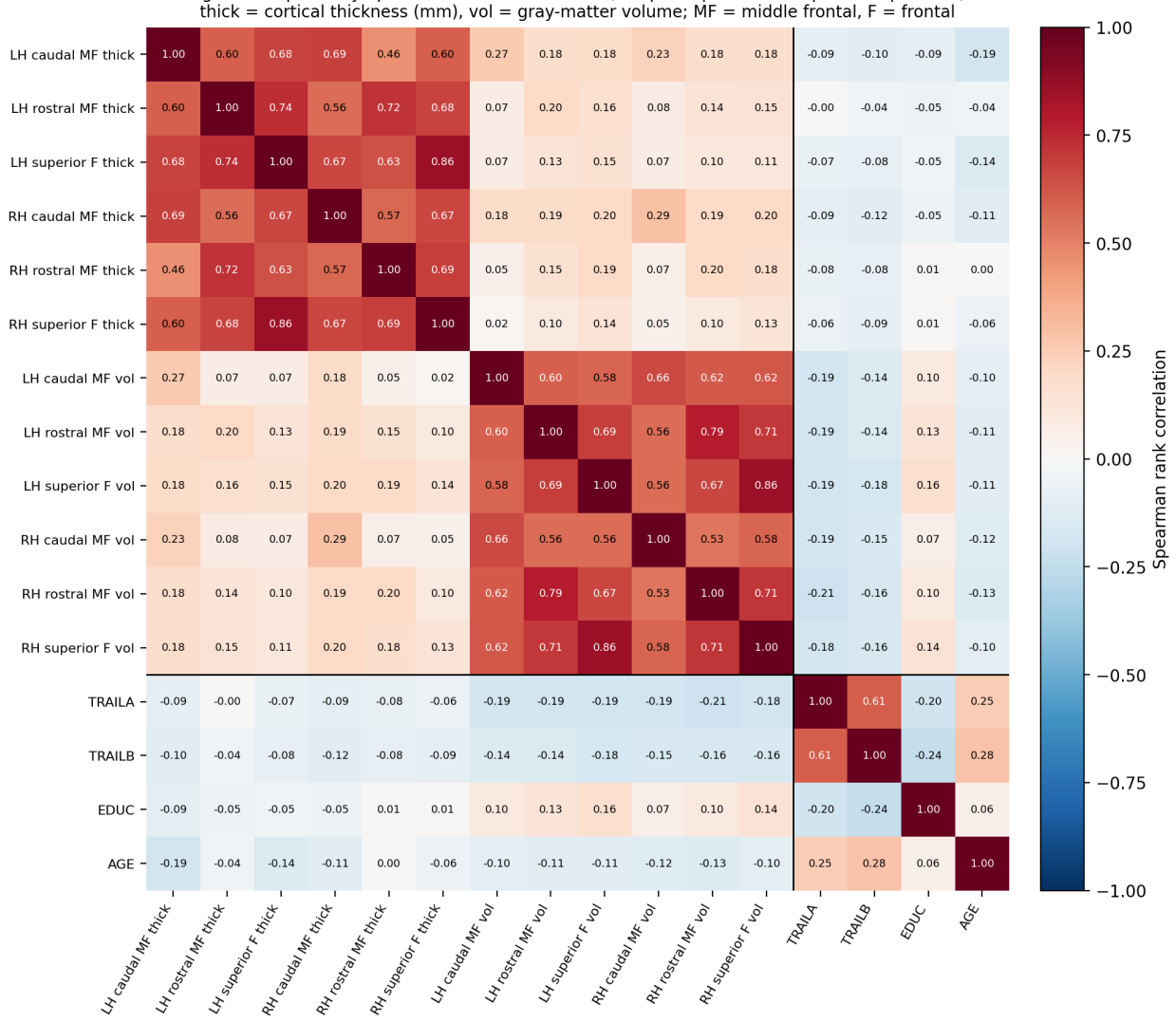


Figure 9. Exploratory correlation matrix of the 12 MRI measures, TRAILA, TRAILB, education and age. Values are descriptive rank correlations.

What the matrix shows.

- MRI measures and TRAILB: all correlations are weak (-0.18 to -0.04). Gray-matter volumes (-0.14 to -0.18) are slightly more strongly related to TRAILB than thickness (-0.04 to -0.12). The sign is the same everywhere: smaller or thinner frontal regions go with a longer (worse) TRAILB time.

- Background variables matter as much as the MRI: TRAILB correlates 0.28 with age and -0.24 with education, and TRAILA and TRAILB correlate 0.61. Volumes also correlate with education (0.07 to 0.16), so part of the volume-TRAILB association may be shared with these variables.

- Strong overlap among MRI measures: thickness measures correlate 0.46-0.86 with one another and volumes 0.53-0.86; left and right values of the same region correlate 0.66-0.86. Entering all 12 as separate predictors in a regression would therefore cause multicollinearity (Phase 4), which supports using a small number of composites.

- Thickness and volume of the same region are only weakly related (0.13-0.29), so they carry different information. Volume also depends on head size, and no intracranial volume is available in the files.

Figure 10. Each frontal MRI measure vs TRAILB (exploratory; Spearman rho shown, no tests)

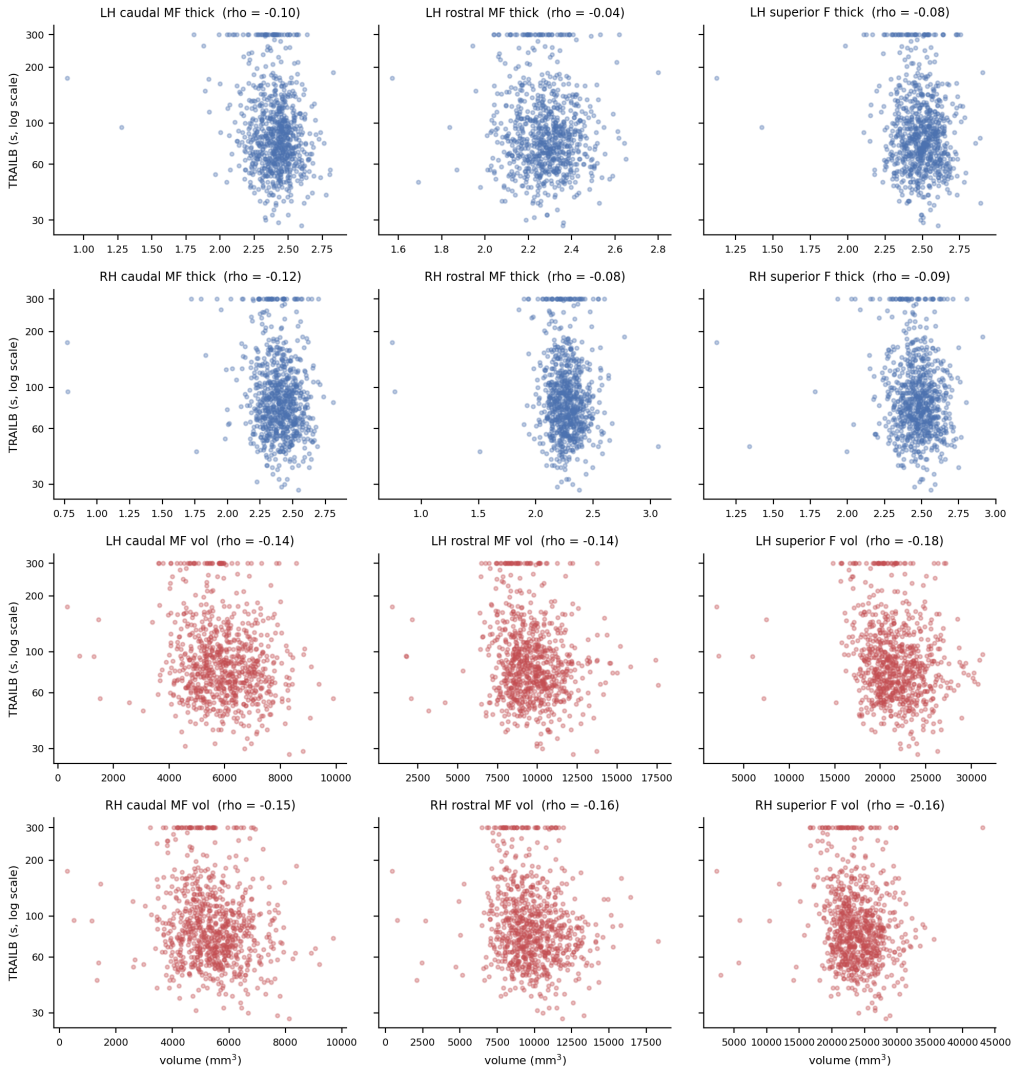


Figure 10. Each frontal MRI measure against TRAILB (rows: left thickness, right thickness, left volume, right volume; columns: three regions). Spearman rho in each panel title; exploratory.

Figure 11. Each frontal MRI measure vs age (exploratory; Spearman rho shown, no tests)

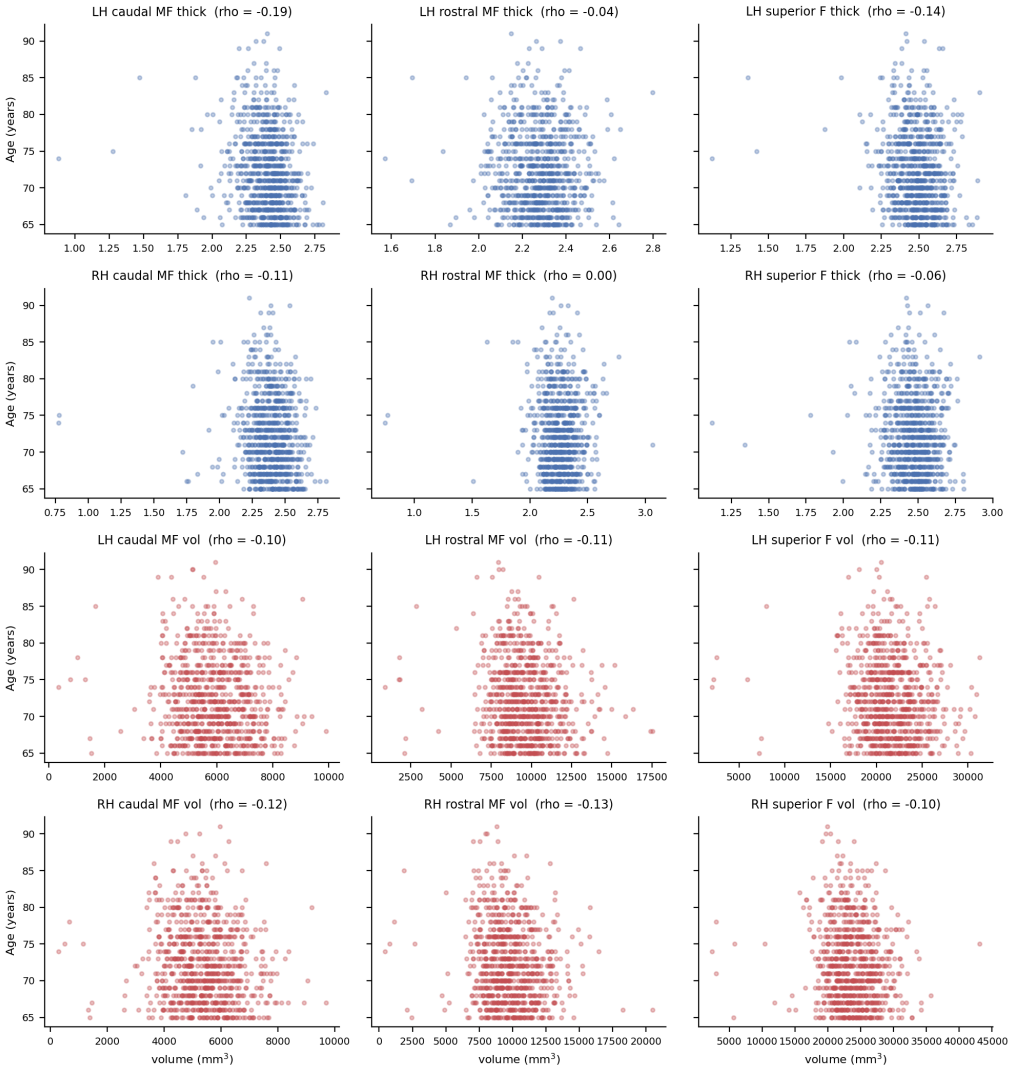


Figure 11. Each frontal MRI measure against age (exploratory).

Figure 12. Thickness-volume and left-right relationships (exploratory)

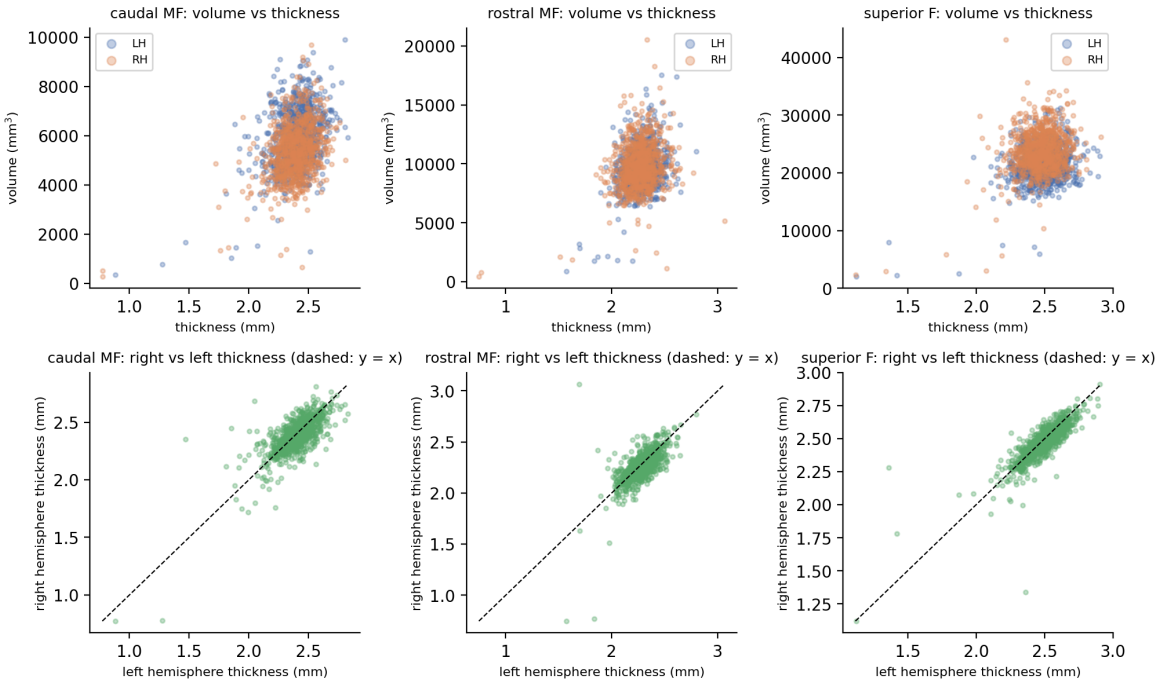


Figure 12. Top: thickness vs volume per region; bottom: right vs left thickness (dashed line y = x).

Figure 13. Frontal MRI measures by living situation (descriptive; Spouse = spouse/partner, Rel/fr = relative/friend)

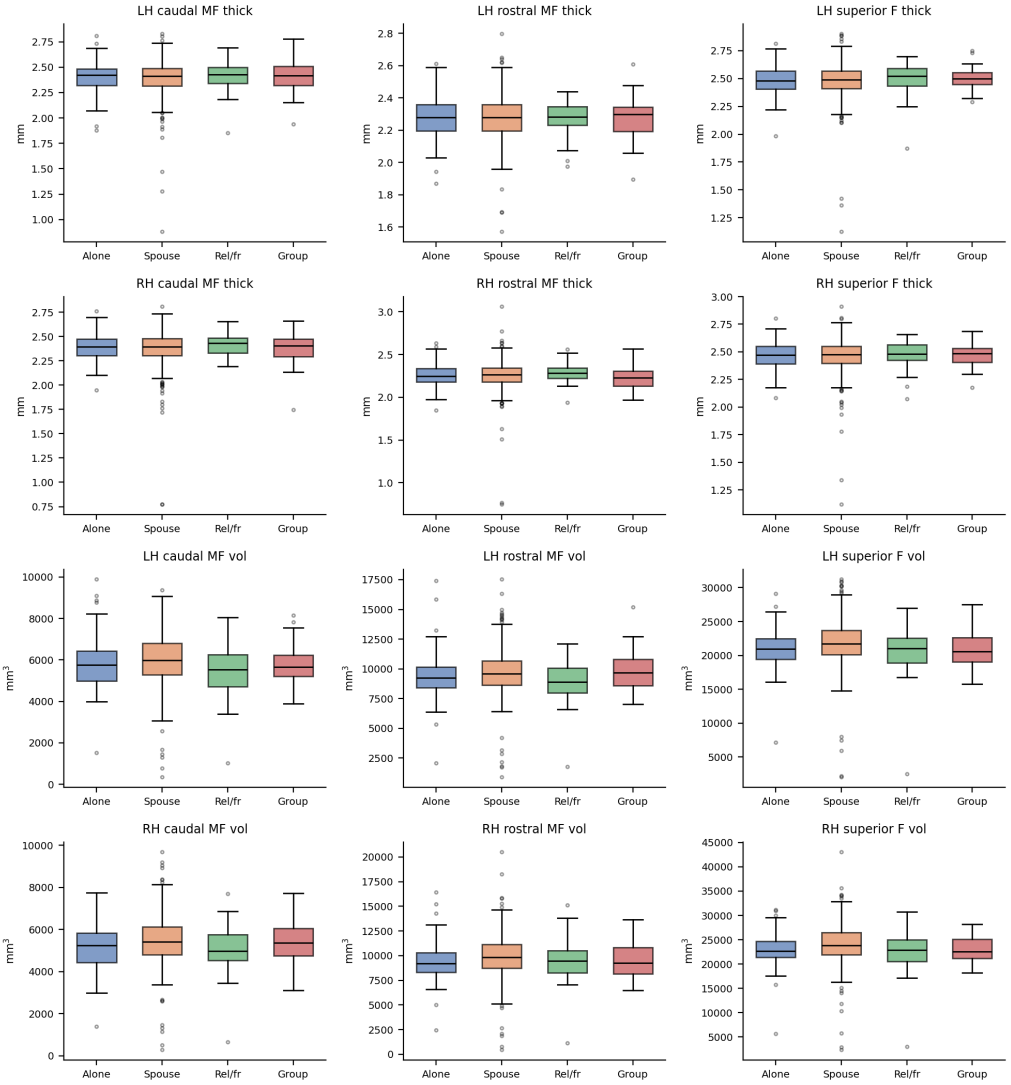


Figure 13. All 12 MRI measures by living situation (alone, spouse/partner, relative/friend, group). Descriptive; the code 5 participant is excluded and code 9 is treated as missing.

MRI measures by living situation. Median thickness is almost identical in the four groups (for example left superior frontal 2.48-2.52 mm). Median volumes differ by up to about 6% between groups (right superior frontal 22,547-23,893 mm^3) without a consistent pattern; with 56 and 36 participants in two groups and no head-size adjustment, these differences should not be over-read. Frontal volumes also contain a few very low values (minimum 277 against a median of roughly 5,400 for caudal middle frontal), probably segmentation errors that will be examined with influence diagnostics in Phases 4-5.

Cautions and consequences. With 16 variables there are 120 pairs; even small correlations can look 'interesting' by chance, and picking the strongest pair and then testing it would be circular. Any later test of these variables will be pre-declared and, if many are tested, adjusted for multiplicity (e.g. FDR). **Decisions:** (1) the confirmatory plan keeps the pre-specified thickness composite and TRAILB; (2) a volume composite of the same six regions is declared now as a Phase 5 sensitivity definition, so that it is not a result-driven choice; (3) living situation is examined across all four groups descriptively, while the primary contrast stays alone versus spouse/partner.

12. Final statistical question and why these data fit (task 10)

Question (unchanged from the project card). Is living situation (NACCLIVS) associated with executive function (TRAILB) and with the cortical thickness of selected frontal regions (mean AVGTH of caudal middle frontal, rostral middle frontal and superior frontal regions) among participants aged 65 or older in the NACC subset?

Precision gained from this phase. (i) Analytic sample A (n = 861: subset participants aged 65+ with valid NACCLIVS 1-4; 864 are 65+ before excluding NACCLIVS 5 and 9); usable-outcome subsamples: TRAILB n = 798, frontal thickness n = 861. (ii) Primary contrast: alone (n = 195) versus with spouse/partner (n = 574); the two other groups (n = 56 and 36) are described and used only in secondary, exploratory comparisons. (iii) Primary TRAILB definition: observed values only (n = 798); the 300 s limit is a censoring problem: a log transformation alone is not enough, so rank-based methods and a sensitivity analysis excluding or censoring the capped values will be considered. (iv) Covariates: age, sex, education. (v) The primary imaging outcome is the six-value frontal composite; individual regions are exploratory.

Sub-questions (so that the question, the analysis and the models match). (1) Does executive function (TRAILB) differ between living-situation groups, accounting for age, sex and education? (2) Does frontal cortical thickness differ between living-situation groups, accounting for the same variables? (3) Complementary, associational: is frontal cortical thickness associated with executive function? **Planned models (Phase 4):** (A) response TRAILB: only-UDS (living situation + age, sex, education), only-MRI (frontal thickness) and combined models; (B) response frontal thickness: living situation + age, sex, education.

Why the data are suitable. Both modalities are available for every participant, the key variables are measured with standard instruments, one record per person guarantees independent observations, and the two largest groups are big enough for a comparison. **Where they are not ideal:** non-random sample, unknown imputation, cross-sectional living situation, small subgroups, and missing TRAILB linked to dementia.

13. Limitations noted in this phase

- **Selection and generalisability:** volunteers in research centres, only participants with MRI, and unknown criteria for building the subset; no statement about older adults in general.

- **Observational data:** living situation is not assigned; reverse causation (cognitive decline changing living arrangements) and confounding by sex, age, health and marital status are plausible.

- **Unknown visit selection:** the delivered record per participant has no visit date; the time gap between UDS and MRI is unknown.

- **Imputation:** 137 columns contain decimal values that suggest prior imputation or transformation; the procedure is unknown. In TRAILB these were excluded from the primary definition.

- **TRAILB:** censored at 300 s, special codes for not completed, and unusable results more common in dementia (44% vs 7.5%).

- **MRI:** no scanner/site or intracranial volume information (so volumes are not head-size adjusted); the frontal composite is an unweighted mean; a few thickness values below 1 mm and very low volumes are probably segmentation errors; the 12 MRI measures are strongly inter-correlated.

- **Coding not fully cross-checked:** SEX, EDUC, MARISTAT, MOCATOTS and NACCAPOE codes and the MRI units are not yet verified against the RDD (Table 6e); they remain a limitation until confirmed.

- **Small groups:** 'relative/friend' (n = 56) and 'group' (n = 36); 'other' (n = 1) excluded.

Open questions for the teaching team (answers will be added to the report):

- Which visit does the single record per participant represent, and how far apart in time are the UDS and MRI data?

- Which imputation or transformation was applied to the UDS file, and for which variables?

- Were the MRI values also processed or replaced? Is the delivered file the final, analysis-ready version?

- Is an intracranial-volume (head-size) measure available, and do the RDD units (AVGTH in mm, GVOL in mm^3) match those provided?

14. Reproducibility

All numbers and figures in this report are produced by `src/run_phase1.py` (same code as `notebooks/phase1_analysis.ipynb`) from the two CSV files placed in `data/`; the report is built by `src/build_report.py`. Environment: Python 3 with pandas, numpy, scipy, matplotlib (see requirements.txt). Fixed seed is not needed in this phase (no randomness). Tables are in `reports/phase1/tables/`, figures in

`figures/`.

References. NACC, About NACC Data, https://www.naccdata.org/about-nacc-data/ (accessed for this report); NACC Researcher's Guide and Data Forms (RDD) for variable definitions; NACC UDS Forms C1/C2 (Neuropsychological Battery), https://files.alz.washington.edu/documentation/uds3-ivp-c2.pdf and NACC UDS derived variables, https://files.alz.washington.edu/documentation/uds2-dervar.pdf; Wonnacott and Wonnacott, *Introductory Statistics*, 5th ed., Wiley.

Appendix A. Variable map of the UDS file (C.1)

Domain and sub-domain were assigned by variable name and must be confirmed in the RDD; 'see RDD' means no description was written. The full table with inferred scale, range and number of distinct values is in `reports/phase1/tables/variable_map_uds.csv`.

variable	domain	subdomain	description
SEX	demographic	demographics	Sex (1 male, 2 female)
HISPANIC	demographic	demographics	Hispanic/Latino ethnicity
HISPOR	demographic	demographics	see RDD
RACE	demographic	demographics	Primary race
PRIMLANG	demographic	demographics	Primary language
EDUC	demographic	demographics	Years of education (99 = unknown)
MARISTAT	demographic	demographics	Marital status
NACCLIVS	demographic	demographics	Living situation (1 alone, 2 spouse/partner, 3 relative/friend, 4 group, 5 other, 9 unknown)
RESIDENC	demographic	demographics	Type of residence
NACCNIHR	demographic	demographics	see RDD
NACCAGE	demographic	demographics	Age at visit (years)
INDEPEND	functional	independence / CDR / FAQ	Level of independence
BILLS	functional	independence / CDR / FAQ	see RDD
TAXES	functional	independence / CDR / FAQ	see RDD
SHOPPING	functional	independence / CDR / FAQ	see RDD
GAMES	functional	independence / CDR / FAQ	see RDD
STOVE	functional	independence / CDR / FAQ	see RDD
MEALPREP	functional	independence / CDR / FAQ	see RDD
EVENTS	functional	independence / CDR / FAQ	see RDD
PAYATTN	functional	independence / CDR / FAQ	see RDD
REMDATES	functional	independence / CDR / FAQ	see RDD
TRAVEL	functional	independence / CDR / FAQ	see RDD
MEMORY	functional	independence / CDR / FAQ	CDR box: memory
ORIENT	functional	independence / CDR / FAQ	CDR box: orientation
JUDGMENT	functional	independence / CDR / FAQ	CDR box: judgment/problem solving
COMMUN	functional	independence / CDR / FAQ	CDR box: community affairs
HOMEHOBB	functional	independence / CDR / FAQ	CDR box: home/hobbies
PERSCARE	functional	independence / CDR / FAQ	CDR box: personal care
CDRSUM	functional	independence / CDR / FAQ	CDR sum of boxes
CDRGLOB	functional	independence / CDR / FAQ	Global CDR
DECSUB	cognitive	neuropsychological tests	Subject reports memory decline
ANIMALS	cognitive	neuropsychological tests	Category fluency: animals
VEG	cognitive	neuropsychological tests	Category fluency: vegetables
TRAILA	cognitive	neuropsychological tests	Trail Making Test Part A, seconds
TRAILARR	cognitive	neuropsychological tests	Trails A: errors (verify in RDD)
TRAILALI	cognitive	neuropsychological tests	Trails A: correct lines (verify in RDD)
TRAILB	cognitive	neuropsychological tests	Trail Making Test Part B, seconds (executive function; higher = worse)
TRAILBRR	cognitive	neuropsychological tests	Trails B: errors (verify in RDD)
TRAILBLI	cognitive	neuropsychological tests	Trails B: correct lines (verify in RDD)
MOCATOTS	cognitive	neuropsychological tests	MoCA total score (0-30)
NACCUDSD	diagnostic	cognitive status	Cognitive status at visit (1 normal, 2 impaired-not-MCI, 3 MCI, 4 dementia)
NACCFAM	medical	family history	Family history of cognitive impairment
NACCMOM	medical	family history	see RDD
NACCDAD	medical	family history	see RDD
NACCAM	medical	family history	see RDD
NACCAMS	medical	family history	see RDD
NACCFM	medical	family history	see RDD
NACCFMS	medical	family history	see RDD

variable	domain	subdomain	description
NACCOM	medical	family history	see RDD
NACCOMS	medical	family history	see RDD
NACCFADM	medical	family history	see RDD
NACCFFTD	medical	family history	see RDD
NACCAPOE	medical	genetic	APOE genotype
ANYMEDS	medical	medications	Taking any medication
NACCAAAS	medical	medications	see RDD
NACCAANX	medical	medications	see RDD
NACCAC	medical	medications	see RDD
NACCACEI	medical	medications	see RDD
NACCADEP	medical	medications	see RDD
NACCAHTN	medical	medications	see RDD
NACCAMD	medical	medications	see RDD
NACCANGI	medical	medications	see RDD
NACCAPSY	medical	medications	see RDD
NACCBETA	medical	medications	see RDD
NACCCCBS	medical	medications	see RDD
NACCDBMD	medical	medications	see RDD
NACCDIUR	medical	medications	see RDD
NACCEMD	medical	medications	see RDD
NACCEPMD	medical	medications	see RDD
NACCHTNC	medical	medications	see RDD
NACCLIPL	medical	medications	see RDD
NACCNSD	medical	medications	see RDD
NACCPDMD	medical	medications	see RDD
NACCVASD	medical	medications	see RDD
TOBAC30	medical	substance use	see RDD
TOBAC100	medical	substance use	see RDD
SMOKYRS	medical	substance use	see RDD
PACKSPER	medical	substance use	see RDD
QUITSMOK	medical	substance use	see RDD
ALCOHOL	medical	substance use	see RDD
ABUSOTHR	medical	substance use	see RDD
CVHATT	medical	cardio/cerebrovascular	see RDD
CVAFIB	medical	cardio/cerebrovascular	see RDD
CVANGIO	medical	cardio/cerebrovascular	see RDD
CVBYPASS	medical	cardio/cerebrovascular	see RDD
CVPACDEF	medical	cardio/cerebrovascular	see RDD
CVCHF	medical	cardio/cerebrovascular	see RDD
CVOTHR	medical	cardio/cerebrovascular	see RDD
CBSTROKE	medical	cardio/cerebrovascular	see RDD
CBTIA	medical	cardio/cerebrovascular	see RDD
HYPERTEN	medical	cardio/cerebrovascular	see RDD
HYPERCHO	medical	cardio/cerebrovascular	see RDD
SEIZURES	medical	neurological	see RDD
NACCTBI	medical	neurological	see RDD
TBI	medical	neurological	see RDD
TBIBRIEF	medical	neurological	see RDD
TBIEXTEN	medical	neurological	see RDD
TBIWOLOS	medical	neurological	see RDD
NACCNREX	medical	neurological	see RDD
NORMEXAM	medical	neurological	see RDD
DIABETES	medical	metabolic / other conditions	see RDD
DIABTYPE	medical	metabolic / other conditions	see RDD
B12DEF	medical	metabolic / other conditions	see RDD

variable	domain	subdomain	description
THYROID	medical	metabolic / other conditions	see RDD
ARTHRIT	medical	metabolic / other conditions	see RDD
ARTHTYPE	medical	metabolic / other conditions	see RDD
ARTHUPEX	medical	metabolic / other conditions	see RDD
ARTHLOEX	medical	metabolic / other conditions	see RDD
ARTHSPIN	medical	metabolic / other conditions	see RDD
ARTHUNK	medical	metabolic / other conditions	see RDD
INCONTU	medical	metabolic / other conditions	see RDD
INCONTF	medical	metabolic / other conditions	see RDD
APNEA	medical	metabolic / other conditions	see RDD
RBD	medical	metabolic / other conditions	see RDD
INSOMN	medical	metabolic / other conditions	see RDD
OTHSLEEP	medical	metabolic / other conditions	see RDD
PTSD	medical	psychiatric history	see RDD
BIPOLAR	medical	psychiatric history	see RDD
SCHIZ	medical	psychiatric history	see RDD
DEP2YRS	medical	psychiatric history	see RDD
DEPOTHR	medical	psychiatric history	see RDD
ANXIETY	medical	psychiatric history	see RDD
OCD	medical	psychiatric history	see RDD
NPSYDEV	medical	psychiatric history	see RDD
PSYCDIS	medical	psychiatric history	see RDD
HEIGHT	medical	physical measures / sensory	Height
WEIGHT	medical	physical measures / sensory	Weight
BPSYS	medical	physical measures / sensory	Systolic blood pressure
BPDIAS	medical	physical measures / sensory	Diastolic blood pressure
HRATE	medical	physical measures / sensory	Resting heart rate
NACCBMI	medical	physical measures / sensory	Body mass index
VISION	medical	physical measures / sensory	see RDD
VISCORR	medical	physical measures / sensory	see RDD
VISWCORR	medical	physical measures / sensory	see RDD
HEARING	medical	physical measures / sensory	see RDD
HEARAID	medical	physical measures / sensory	see RDD
HEARWAID	medical	physical measures / sensory	see RDD
NPIQINF	medical	behavioural symptoms (NPI-Q)	see RDD
DEL	medical	behavioural symptoms (NPI-Q)	see RDD
DELSEV	medical	behavioural symptoms (NPI-Q)	see RDD
HALL	medical	behavioural symptoms (NPI-Q)	see RDD
HALLSEV	medical	behavioural symptoms (NPI-Q)	see RDD
AGIT	medical	behavioural symptoms (NPI-Q)	see RDD
AGITSEV	medical	behavioural symptoms (NPI-Q)	see RDD
DEPD	medical	behavioural symptoms (NPI-Q)	see RDD
DEPDSEV	medical	behavioural symptoms (NPI-Q)	see RDD
ANX	medical	behavioural symptoms (NPI-Q)	see RDD
ANXSEV	medical	behavioural symptoms (NPI-Q)	see RDD
ELAT	medical	behavioural symptoms (NPI-Q)	see RDD
ELATSEV	medical	behavioural symptoms (NPI-Q)	see RDD
APA	medical	behavioural symptoms (NPI-Q)	see RDD
APASEV	medical	behavioural symptoms (NPI-Q)	see RDD
DISN	medical	behavioural symptoms (NPI-Q)	see RDD
DISNSEV	medical	behavioural symptoms (NPI-Q)	see RDD
IRR	medical	behavioural symptoms (NPI-Q)	see RDD
IRRSEV	medical	behavioural symptoms (NPI-Q)	see RDD
MOT	medical	behavioural symptoms (NPI-Q)	see RDD
MOTSEV	medical	behavioural symptoms (NPI-Q)	see RDD

variable	domain	subdomain	description
NITE	medical	behavioural symptoms (NPI-Q)	see RDD
NITESEV	medical	behavioural symptoms (NPI-Q)	see RDD
APP	medical	behavioural symptoms (NPI-Q)	see RDD
APPSEV	medical	behavioural symptoms (NPI-Q)	see RDD
NOGDS	medical	mood (GDS)	see RDD
SATIS	medical	mood (GDS)	see RDD
DROPACT	medical	mood (GDS)	see RDD
EMPTY	medical	mood (GDS)	see RDD
BORED	medical	mood (GDS)	see RDD
SPIRITS	medical	mood (GDS)	see RDD
AFRAID	medical	mood (GDS)	see RDD
HAPPY	medical	mood (GDS)	see RDD
HELPLESS	medical	mood (GDS)	see RDD
STAYHOME	medical	mood (GDS)	see RDD
MEMPROB	medical	mood (GDS)	see RDD
WONDRFUL	medical	mood (GDS)	see RDD
WRTHLESS	medical	mood (GDS)	see RDD
ENERGY	medical	mood (GDS)	see RDD
HOPELESS	medical	mood (GDS)	see RDD
BETTER	medical	mood (GDS)	see RDD
NACCGDS	medical	mood (GDS)	Geriatric Depression Scale total

Table A1. UDS variables (179) with domain, sub-domain and description.

Appendix B. Variable map of the MRI file (C.1)

lobe	regions (each has LH/RH x GVOL/AVGTH)
cingulate	CAUDALANTERIORCINGULATE, ISTHMUSCINGULATE, POSTERIORCINGULATE, ROSTRALANTERIORCINGULATE
frontal	CAUDALMIDDLEFRONTAL, LATERALORBITOFRONTAL, MEDIALORBITOFRONTAL, PARACENTRAL, PARSOPERCULARIS, PARSORBITALIS, PARSTRIANGULARIS, PRECENTRAL, ROSTRALMIDDLEFRONTAL, SUPERIORFRONTAL
insula	INSULA
occipital	CUNEUS, LATERALOCCIPITAL, LINGUAL, PERICALCARINE
parietal	INFERIORPARIETAL, POSTCENTRAL, PRECUNEUS, SUPERIORPARIETAL, SUPRAMARGINAL
temporal	ENTORHINAL, FUSIFORM, INFERIORTEMPORAL, MIDDLETEMPORAL, PARAHIPPOCAMPAL, SUPERIORTEMPORAL, TRANSVERSETEMPORAL

Table B1. The 31 cortical regions by lobe. GVOL = gray-matter volume (mm^3); AVGTH = average cortical thickness (mm); units as provided for this data set, to be cross-checked against the RDD. Column names: {LH|RH}_{REGION}_{GVOL|AVGTH}.